Data Visualization Mini Project - Global Food Wastage

Project Objectives

This project aims to assess the magnitude of global food waste and its associated economic losses over recent years, and to generate dat a-driven insights that identify the relative impact of contributing factors and inform future policy priorities.

The hypotheses guiding this project are as follows:

1. Global economic and population growth contribute to increased food demand, which in turn drives overproduction and exacerbates foo d waste.

2. Developed countries experience greater economic losses from food waste compared to developing nations.

Dataset Description

The dataset [1], comprising 5,000 records, provides comprehensive information on total food waste (measured in tons), average per capi ta waste (in tons), and associated economic losses (in millions of dollars), categorized by eight food types across 20 countries from 2018 t o 2024. Appendix 1 provides a detailed overview of the dataset's characteristics.

Methodology

This project utilizes Tableau as the core tool for developing interactive and analytical data visualizations.

Selecting relevant visualization from Tableau:

Type of Visualization	Description
Map	Geospatial data (longitude and latitude) contained in the dataset is utilized within Tableau’ s ma p view to effectively illustrate the geographic distribution of food waste and associated economic losses across 20 countries.
Lines Chart	Since the dataset includes longitudinal data across multiple years, a line chart is employed to visu alize trends over time using multiple lines. For example, it is used to compare the economic loss associated with eight food categories from 2018 to 2024 (Appx. 3, Graph 14), providing a clear v iew of category-specific changes and patterns.
Heatmap	The heatmap enhances the visual overview of total food waste across 20 countries and eight food categories by applying a green-to-yellow gradient palette. Grids shaded in yellow or light green i ndicate higher quantities of food waste for the corresponding country and category, while darker green tones represent lower levels. With the year filter applied, changes in total food waste over t ime are clearly observable across all countries.
Bar-Line Chart	A combined bar-line chart is utilized to visualize the relationship between total food waste and ec onomic loss over time, enabling clear comparison across multiple years.
Treemap	By encoding data through proportional rectangles and gradient color scales, the treemap chart pro

	vides a clear representation of economic loss from food waste across 20 countries, highlighting di sparities among categories and regions.
Scatter Plot	A scatter plot is utilized to explore the relationship between two quantitative variables, segmented by a categorical dimension. For instance, to examine the impact of population size on food waste, the chart compares total food waste and average per capita food waste across countries, allowing for clearer insights into national-level patterns.
Donut Chart	A donut chart is employed to visualize the percentage of economic loss attributed to each specific food category. With year and country filters applied, the chart provides a clear and focused repres entation of the distribution of economic loss for a selected country or time period.
Radar Chart	Radar charts are used for comparing multiple variables at once, making it easy to spot patterns, di fferences, and outliers across categories. They clearly show how data sets like Spain’ s and Sout h Korea’ s economic loss profiles differ, highlighting strengths and weaknesses in a simple and visual way.
Rank Chart	A rank chart is used to present the ranking of categorical data based on associated quantitative va lues. In this context, it displays the ranking of food waste categories over time across different co untries. The chart also offers insights into national food consumption habits, reflecting how prefer ences and waste patterns vary by region.

Applying Functions and Interactive Features from Tableau:

Features	Description
Filter	Year, country, and food category are utilized as interactive filters to enable multidimensional vis ual comparisons across the dataset.
Highlighter	Highlighters were used for country and food categories, which helps emphasize specific data poi nts, making it easier to compare contributions and performance across countries and food categor ies.
Marks (Color, Size, Label, Detail, Tooltip)	Color encoding is applied to distinguish the countries by development status—developed or deve loping—while a green-to-yellow gradient scale is used to visualize the magnitude of total food w aste and economic loss. Additionally, distinct colors are assigned to each of the eight food catego ries, enhancing clarity and enabling quick visual differentiation across types of food waste. Size is used to depict economic loss from food waste in 20 countries in a treemap, where the area of each rectangle reflects the corresponding value.
Set	Based on United Nations classifications [3], the 20 countries in this study were grouped into 9 de veloped and 11 developing categories by creating a set in Tableau. This categorization enables co mparative analysis of food waste across countries according to their level of economic developm ent.
Calculated Field	Calculated Fields e.g. Remainder of Economic Loss in each Category was used in the Donut Cha rt for better visualization.

Highlights

(1) Comprehensive Analysis of Global Food Waste: Economic, Demographic, Category, and Temporal Perspectives

The map (Appx. 3, Graph 2) visualizes total food waste (in tons) from 2018 to 2024 using a green-to-yellow gradient scale. Countries with the highest levels of food waste—such as Turkey, Canada, Spain, and Germany—are highlighted in yellow or greenish-yellow, indicating greater severity. In contrast, South Korea, Australia, and Japan appear in varying shades of green, suggesting a lower contribution to food waste among the 20 countries analyzed. By adjusting the year filter, temporal changes become evident; however, the overall quantity of food waste remains substantial across all countries throughout the period.

As shown in the scatter plot (Appx. 3, Graph 3), in addition to highlighting the severity of total food waste across countries, the chart also illustrates the relationship between total food waste and average food waste per capita. Notably, Germany ranks fourth in total food waste but leads in average food waste per capita, indicating a disproportionate contribution relative to its population size. In contrast, the United Kingdom records the lowest overall total food waste and average per capita waste from 2018 to 2024, though it is not consistently ranked lowest each year. The scatter plot reveals a proportional relationship between total and per capita food waste; however, the data points are loosely clustered around the trend line, suggesting that population size is not a decisive factor in determining total food waste.

In comparing developing and developed countries, the line chart (Appx. 3, Graph 4) reveals that developing countries consistently exhibit higher average food waste per capita from 2018 to 2024. The scatter plot (Appx. 3, Graph 5) further illustrates the distribution of household waste as a percentage of total food waste across 20 countries, with blue representing developing countries and orange representing developed ones. The visualization shows that developing countries tend to have a higher proportion of household waste, forming subtle clusters of blue and orange data points. This pattern suggests that, compared to developing countries, developed nations may benefit from more advanced recycling infrastructure and greater public awareness around food conservation and recycling—contributing to their lower average food waste per capita and reduced household waste percentages.

The heatmap (Appx. 3, Graph 6), line chart (Appx. 3, Graph 7), and rank chart (Appx. 3, Graph 8) are utilized to analyze trends in food waste by category and country over time. In the heatmap, countries are vertically ranked in ascending order based on total food waste, with food categories similarly ordered. Consistent with the approach used in the map visualization, the heatmap applies a green-to-yellow gradient to indicate severity. The analysis reveals that prepared food, beverages, and bakery items are the most common contributors to food waste globally—reflecting widespread consumption and resulting in substantial waste volumes. Yellow-shaded grids signal areas of concern. For example, Mexico, Brazil, and Spain are among the top contributors to prepared food waste. Among the three categories mentioned, prepared food waste shows a clear upward trend over the seven years, highlighting an increasingly urgent issue. While Japan ranks among the lowest in total food waste overall, it generates a disproportionately high volume of beverage waste compared to other categories. This finding underscores the need for targeted policies in Japan to curb beverage overproduction and promote more effective recycling practices.

(2) Impact of Food Waste on Economic Loss Across Countries and Development Levels

The barline chart (Appx. 3, Graph 9) illustrates a strong positive correlation between economic loss and total food waste. While no significant increases or decreases are observed, a slight upward trend is apparent, along with noticeable differences between 2018 and 2024, suggesting that this is a persistent and ongoing issue.

The distribution of economic loss by country was examined using a map (Appx. 3, Graph 10) and tree map (Appx. 3, Graph 11) to visually highlight regional differences and proportions of food waste-related economic losses. It was found that these countries all suffer from substantial economic losses, indicating that food waste is not concentrated in a few regions but is a global issue. Clearly, Spain, Turkey, Canada, and Germany experience the highest economic losses, suggesting a need for stronger food supply chain management and waste reduction policies in these countries. Although countries such as South Korea, the UK, and Russia show relatively lower losses, they still face significant economic impacts, further demonstrating that food waste remains a global challenge. Additionally, European countries—including Spain, Germany, France, Italy, and the UK—are prominently represented at both the high and low ends of the loss spectrum, highlighting considerable variability within the region.

As shown in the line chart (Appx. 3, Graph 12), the comparison between developing and developed countries was further analyzed. It w

as observed that both economic loss and total food waste are higher in developing countries than in developed ones. When comparing the two groups, similar trends are observed, showing that food waste negatively impacts economic efficiency regardless of a country's development status. After 2023, a slight decline is noticeable in developed countries, suggesting possible progress in waste reduction and improvements in economic efficiency.

(3) Impact of Key Food Categories on Economic Loss Across Countries: Comparative Trends and Profiles

The Donut Chart (Appx. 3, Graph 13) shows that prepared foods, beverages, and fruits and vegetables are the top contributors to economic loss. The lines frequently intersect in the line chart (Appx. 3, Graph 14), showing large year-to-year fluctuations with no consistent ranking pattern. However, Prepared Food consistently ranks among the top three categories, causing economic loss to steadily increase after 2020 and peaking in 2023. In 2024, its waste amount is noticeably higher than all other categories. In contrast, Grains & Cereals and Bakery Items show more erratic trends, with Grains & Cereals generally low but peaking around 2021, and Bakery Items dropping sharply in 2021 before rising again through 2023. Overall, while most categories' rankings shift over time, Prepared Food consistently contributes a high level of waste, highlighting it as a key area for food waste reduction efforts.

The Scatter Plot (Appx. 3, Graph 15) illustrates the relationship between economic loss and total food waste across different food categories in various countries, clearly showing the economic loss attributed to each food category within each country. The Radar chart (Appx. 3, Graph 16) reveals the economic loss proportions of different food categories in each country, allowing further comparison of their distribution and how their economic loss profiles differ. A comparison between Spain, which has the highest economic loss, and South Korea, with lower losses, reveals distinct patterns. South Korea's losses are relatively balanced across categories, without extreme spikes, and categories like Frozen Food, Dairy Products, and Grains & Cereals show values closer to Spain. In contrast, Spain's losses are skewed, with very large losses concentrated in Prepared Food and Meat & Seafood. Prepared Food represents the largest gap between the two countries, with Spain's economic loss far exceeding South Korea's. Meat & Seafood losses are also significantly higher in Spain, while South Korea's losses in this category are relatively small. This suggests Spain needs urgent interventions to reduce waste in Prepared Food and Meat & Seafood, whereas South Korea should focus on Fruits & Vegetables, Frozen Food, and Beverages, where its losses are higher. Overall, Spain faces larger total food-related economic losses, while South Korea's losses are more evenly spread across multiple categories.

Effectiveness of the visualizations

Tableau's automated aggregation and gradient-based labeling effectively highlight the increasing severity of global food waste and economic loss between 2018 and 2024 (Appx. 3, Graph 1), thereby supporting the study's first hypothesis.

Regarding the second hypothesis—which assumes that developed countries experience greater economic loss from food waste than developing countries due to higher living costs—the data tells a different story. A comparative line chart (Appx. 3, Graph 12) reveals a strong association between the quantity of food waste and economic loss in both groups, suggesting that differences in food cost levels are not a primary contributing factor. Furthermore, developing countries consistently exhibit higher average food waste per capita from 2018 to 2024 (Appx. 3, Graph 4). This trend implies that developing countries may, in fact, suffer greater economic loss from food waste, contradicting the second hypothesis.

Additional ideas or approaches that were not implemented

This study aimed to conduct a comprehensive analysis of global food waste, extending beyond the dimensions provided in the original dataset to include additional variables outlined below. However, the dataset—compiled by the author on Kaggle—was aggregated from multiple sources. Due to concerns about potential inconsistencies that could arise from integrating external datasets using our methodology, this approach was ultimately discarded.

Several exploratory ideas were also considered but not implemented due to analytical or technical limitations:

1. When comparing household waste, economic loss, and food categories, no clear trend or relationship was observed; therefore, this an alysis is not included in this study.

2. Creating a Radar Chart using calculated fields is complicated so we chose to use a Radar Chart from the Viz Extensions instead.

Challenges and Limitations

The dataset includes variables across food waste amounts, food categories, and population size, but several key dimensions—such as fo od pricing, recycling infrastructure, and consumer behavior—were not available.

As the dataset contains only annual data, it limits the ability to capture seasonal patterns or short-term fluctuations resulting from global or regional events.

Future Work - Potential areas for future improvement

Future studies could improve insights by incorporating variables such as GDP per capita to reflect economic context, considering urbani zation rates and their impact on consumption/waste, assessing food supply chain efficiency, and integrating infrastructure development in dicators.

Moreover, an additional dataset containing food production, import, and export volumes is required to assess the food demand and sever ity of overproduction. These variables are also essential for enabling fair comparisons of food waste across different stages—production, supply chain, and retail—between developed and developing countries.

Conclusion

At the country level, the top three countries with the highest food waste and economic loss include developing countries Spain and Turk ey, alongside developed country Canada. Developing countries consistently show higher economic losses and total food waste, while dev eloped countries tend to have lower household food waste and lower food waste per capita, likely due to greater awareness, education, an d recycling infrastructure. Population size does not show a strong correlation with food waste levels, indicating that higher consumption d oes not necessarily lead to more waste.

At the food category level, prepared food stands out as a leading contributor to global food waste and economic loss, making it a crucial area to target for reduction efforts.

Overall, there is a strong positive correlation between the quantity of food waste and economic loss, highlighting a significant global ch allenge. Food waste carries not only environmental but also substantial economic costs worldwide year after year.

Reference

1. https://www.kaggle.com/datasets/atharvasoundankar/global-food-wastage-dataset-2018-2024/data

2. https://wedocs.unep.org/handle/20.500.11822/45230

3. https://www.un.org/en/development/desa/policy/wesp/wesp_current/2014wesp_country_classification.pdf

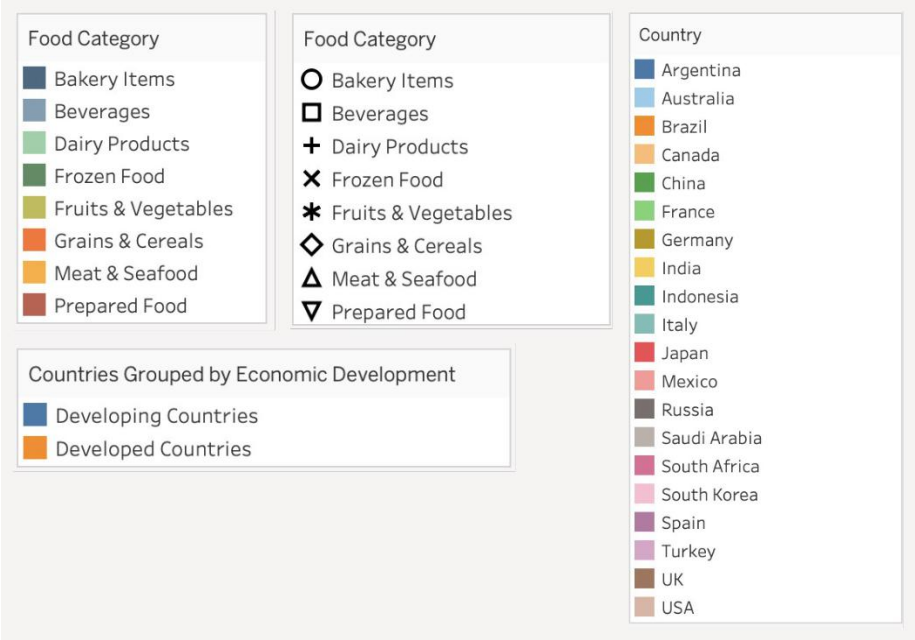
Appendix

Appendix 1: Data Characteristics

No.	Variables	Description	Variable Categories	Variable Type
1	Country	Specifies the country associated with each record. 20 countries are included: - Argentina - Australia - Brazil - Canada - China - France - Germany - India - Indonesia - Italy - Japan - Mexico - Russia - Saudi Arabia - South Africa - South Korea - Spain - Turkey - United Kingdom - United States	Qualitative	Nominal
2	Year	Indicates the calendar year associated with each record, ranging from 2018 to 2024	Quantitative	Discrete
3	Food Category	Denotes one of eight food types used to classify waste data. - Bakery Items - Beverages - Dairy Products - Frozen Food - Fruits & Vegetables - Grains & Cereals - Meat & Seafood - Prepared Food	Qualitative	Nominal
4	Total Waste (Tons)	Represents the total volume of food waste in tons for the associated country, category, and year.	Quantitative	Continuous

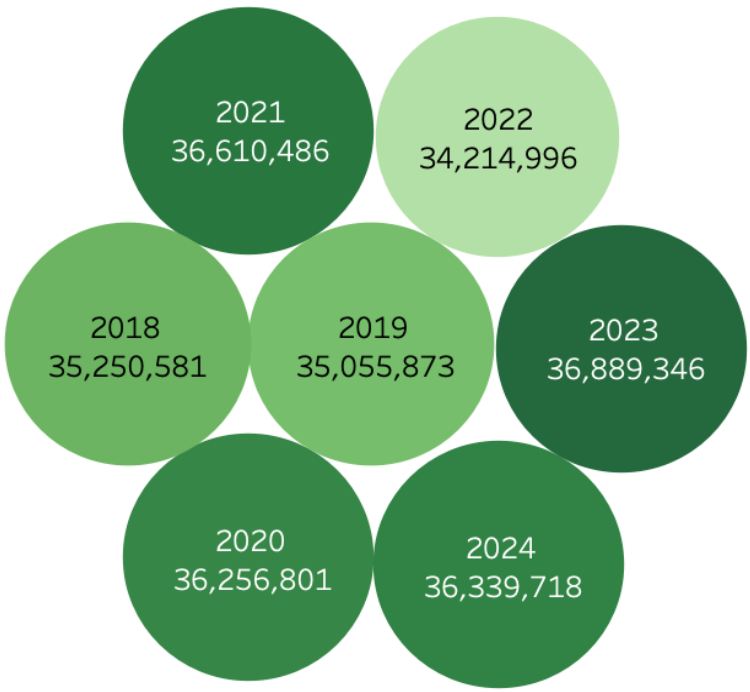
5	Economic Loss (Million $)	Quantifies the financial loss resulting from food waste for th e associated country and food category, expressed in million s of dollars.	Quantitative	Continuous
6	Avg Waste per Capita (Kg)	Measures the average food waste per person in kg.	Quantitative	Continuous
7	Population (Million)	Represents the population in millions for the associated cou ntry and year.	Quantitative	Continuous
8	Household Waste (%)	Represents the percentage of household waste for the associ ated country, category, and year.	Quantitative	Continuous

Appendix 2: Label descriptions for the dataset variables used in the analysis



Appendix 3: Visualisation in Tableau

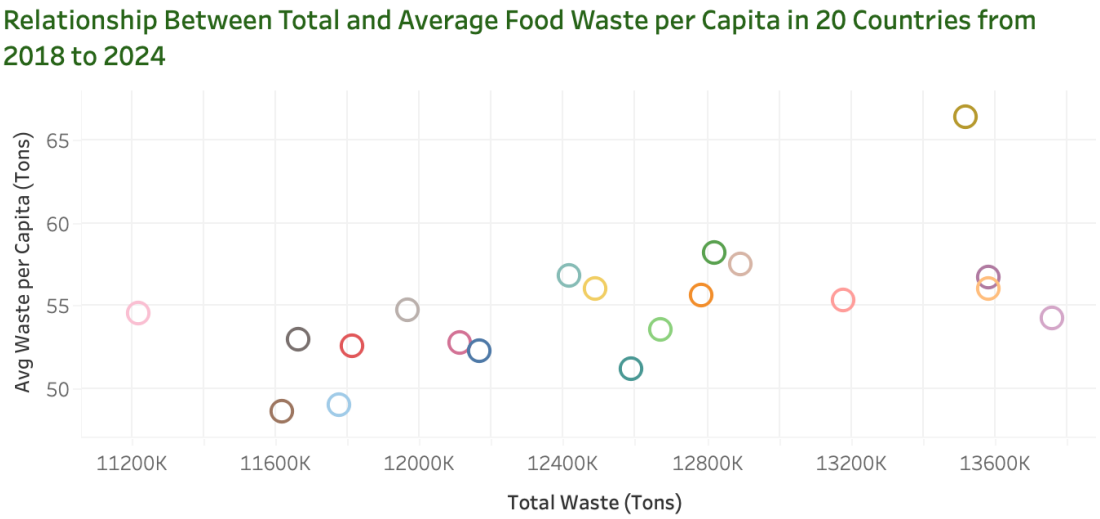
Graph 1: Sum of Food Waste (tons) in 20 Countries from 2018 to 2024



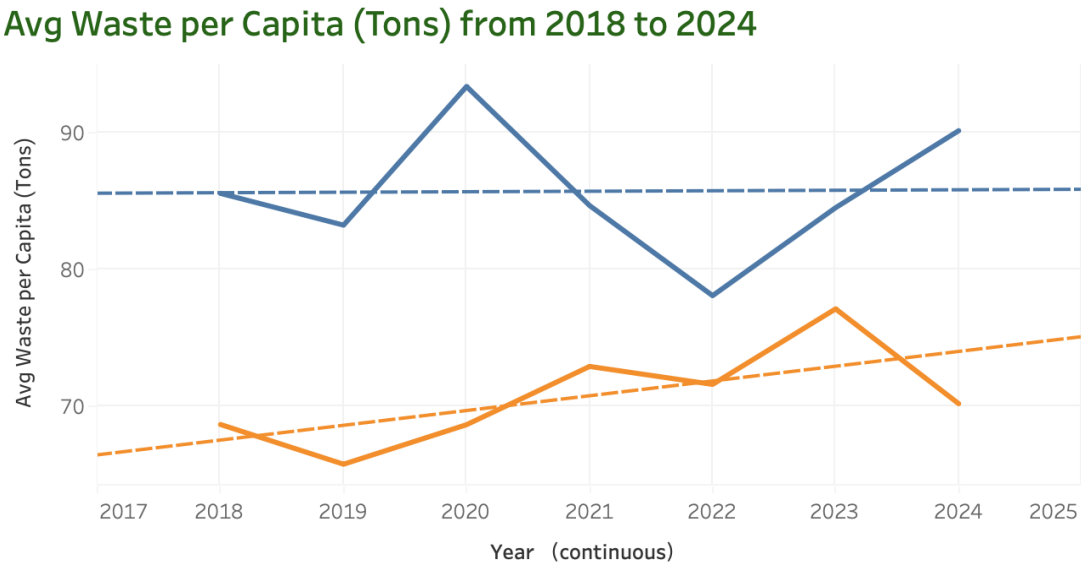
Graph 2: Map - Total Food Waste in 20 Countries from 2018 to 2024



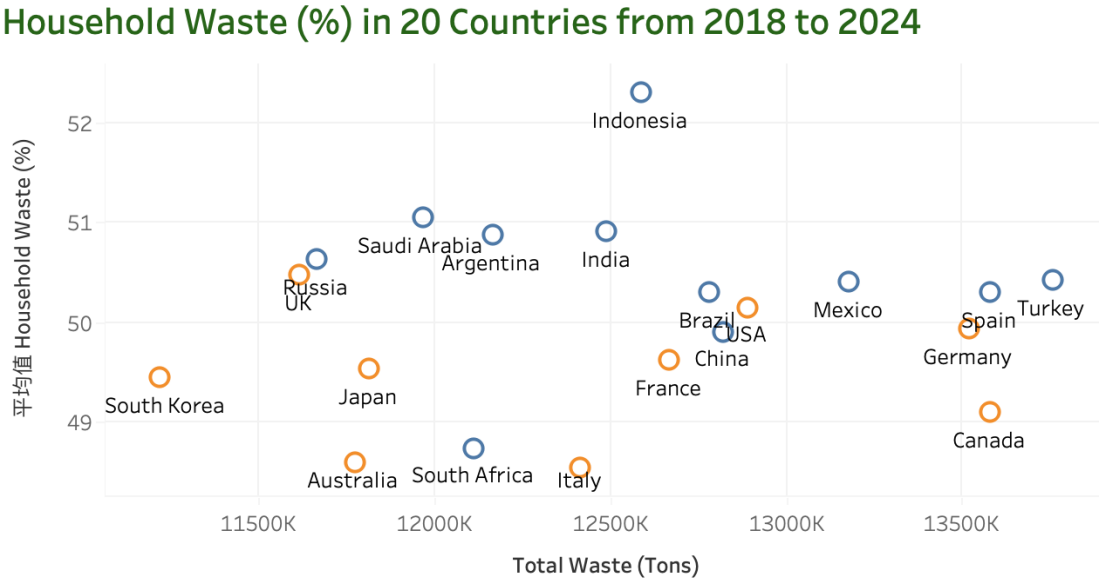
Graph 3: Scatter - Relationship Between Total and Average Food Waste per Capita in 20 Countries from 2018 to 2024



Graph 4: Line - Average Waste per Capita (Tons) from 2018 to 2024



Graph 5: Scatter - Household Waste (%) by 20 Countries from 2018 to 2024

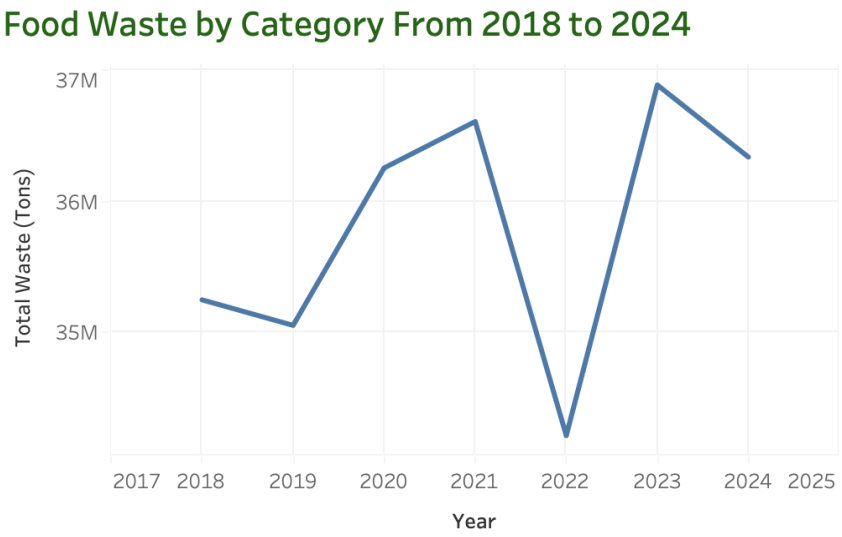


Graph 6: Heatmap - Quantities of Food Waste by Category in 20 Countries (Tons)

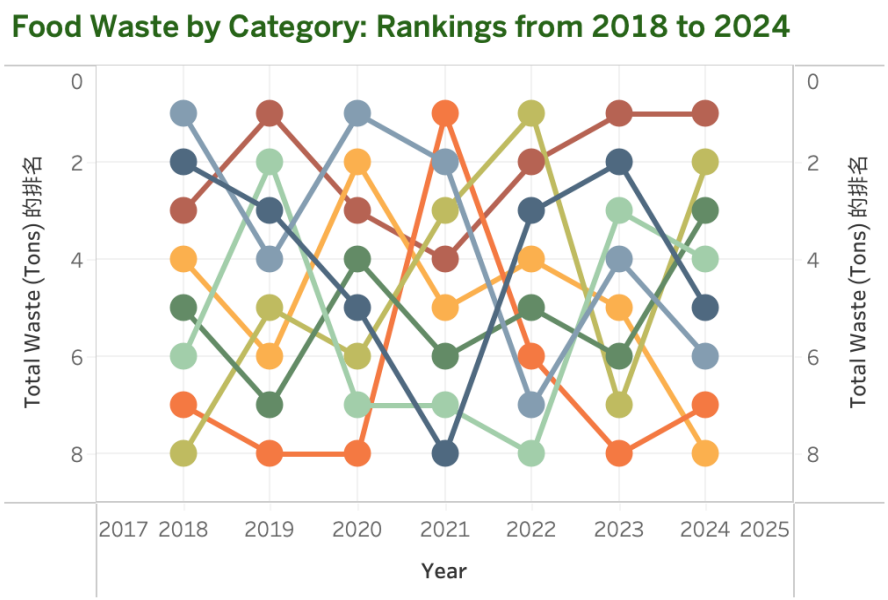
Quantities of Food Waste by Category in 20 Countries (Tons)

Count..	Food Category: Prepared Food	Beverages	Bakery Items	Fruits & Vegetables	Meat & Seafood	Dairy Products	Frozen Food	Grains & Cereals
Turkey	1,829,098	1,955,806	1,663,705	1,723,435	1,901,021	1,163,375	1,640,661	1,882,670
Canada	1,619,372	1,599,877	2,290,236	1,326,063	2,036,506	2,211,067	1,017,580	1,480,656
Spain	2,303,549	1,556,891	1,470,465	1,644,610	1,651,507	1,419,061	1,893,969	1,641,097
Germany	2,086,158	2,056,772	1,604,338	1,890,230	1,139,211	1,512,189	1,641,979	1,589,229
Mexico	2,410,884	1,651,787	1,385,071	1,290,368	1,809,087	1,563,393	1,547,243	1,520,507
USA	1,464,511	1,146,765	2,114,874	1,826,278	1,749,796	1,855,716	1,611,367	1,123,362
China	1,595,249	1,234,940	1,437,945	1,448,477	1,816,705	1,732,365	1,845,810	1,710,702
Brazil	2,398,633	1,505,051	1,606,382	1,607,348	1,304,765	1,357,256	1,468,698	1,535,372
France	1,689,105	1,705,808	1,261,387	1,320,770	1,592,525	1,920,721	1,763,547	1,415,127
Indonesia	1,451,040	2,216,273	1,517,217	1,493,777	1,620,771	1,721,566	1,483,697	1,084,554
India	1,467,473	2,239,700	1,618,023	1,735,840	1,375,709	1,328,107	1,669,319	1,056,457
Italy	2,029,617	1,330,074	1,492,481	1,507,771	1,959,421	1,802,598	814,556	1,478,896
Argentina	1,866,744	1,251,072	1,709,312	2,059,171	1,612,078	883,980	1,643,455	1,138,925
South Africa	1,301,923	1,857,505	1,622,870	1,764,130	1,562,115	990,395	1,212,098	1,801,905
Saudi Arabia	1,997,777	1,463,021	1,634,938	1,071,043	1,427,544	1,865,586	1,208,731	1,297,927
Japan	1,605,184	2,208,988	1,343,681	1,127,808	1,413,006	1,219,164	1,395,861	1,501,291
Australia	1,595,040	1,178,352	1,584,538	1,677,397	1,182,229	1,589,784	1,411,831	1,555,552
Russia	1,700,210	1,509,891	1,641,190	1,477,908	1,859,646	1,329,269	1,139,928	1,007,337
UK	1,979,214	1,485,367	1,165,167	1,286,885	1,076,631	1,658,190	1,863,832	1,101,957
South Korea	1,467,855	1,554,972	1,012,090	1,764,853	652,422	1,460,555	1,816,069	1,489,398

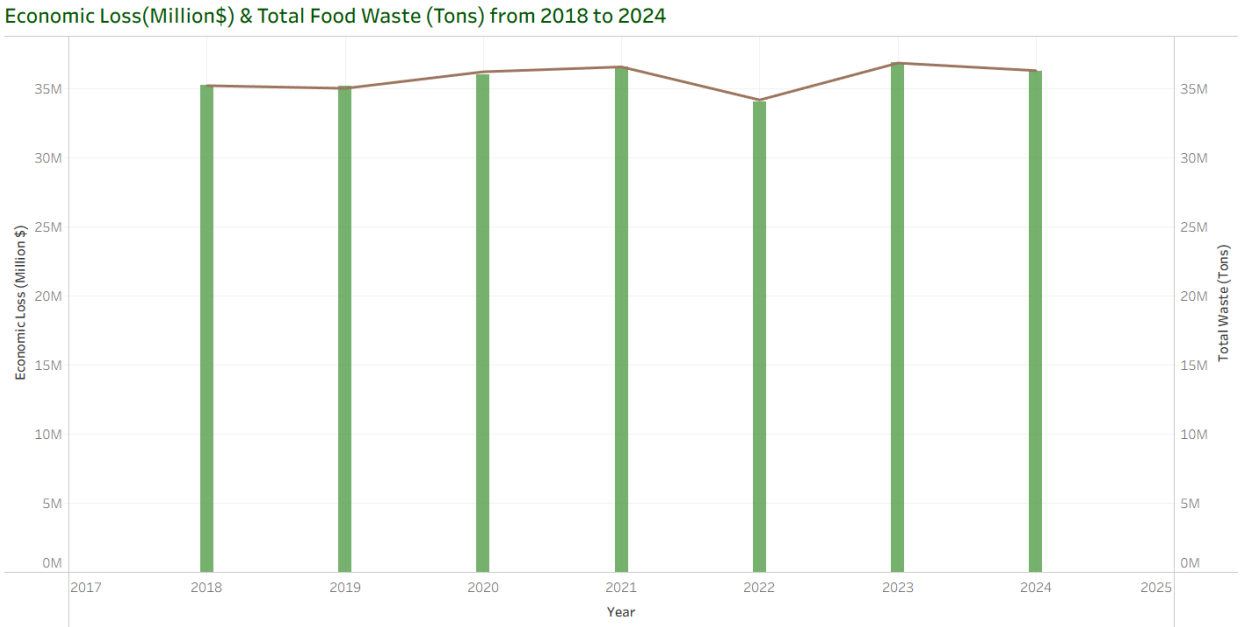
Graph 7: Line - Food Waste by Category From 2018 to 2024



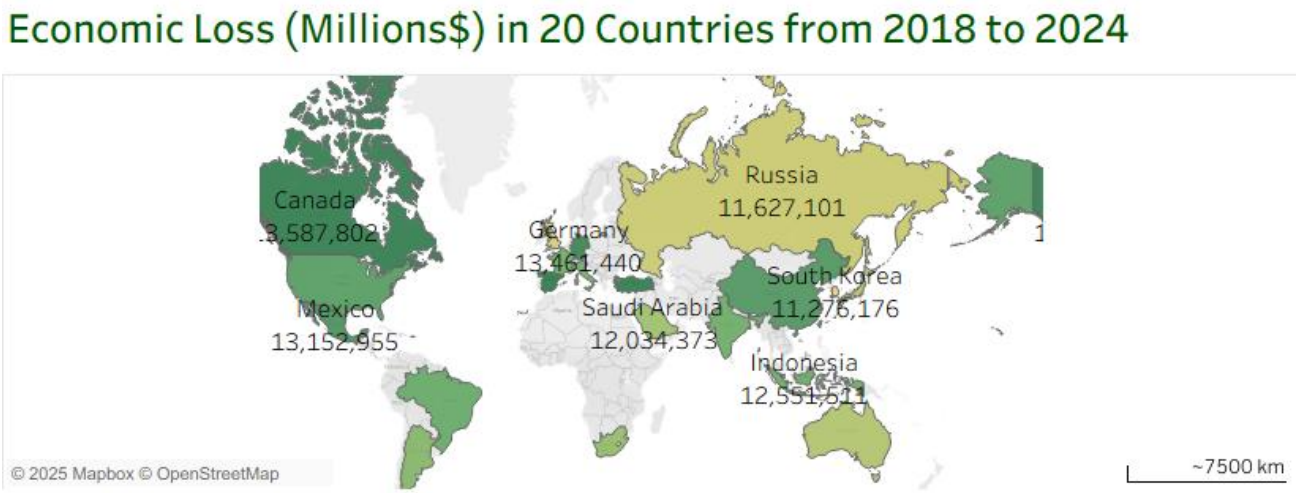
Graph 8: Rank - Food Waste by Category: Rankings from 2018 to 2024



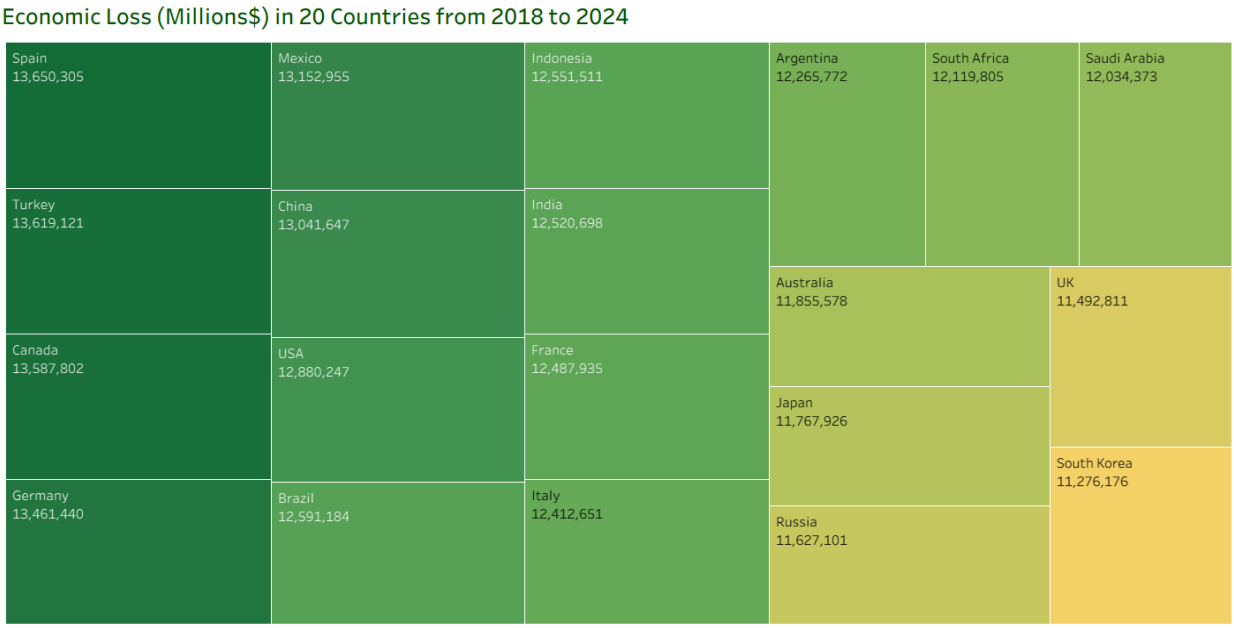
Graph 9: Barline - Economic Loss(Million$) vs Total Food Waste (Tons) from 2018 to 2024



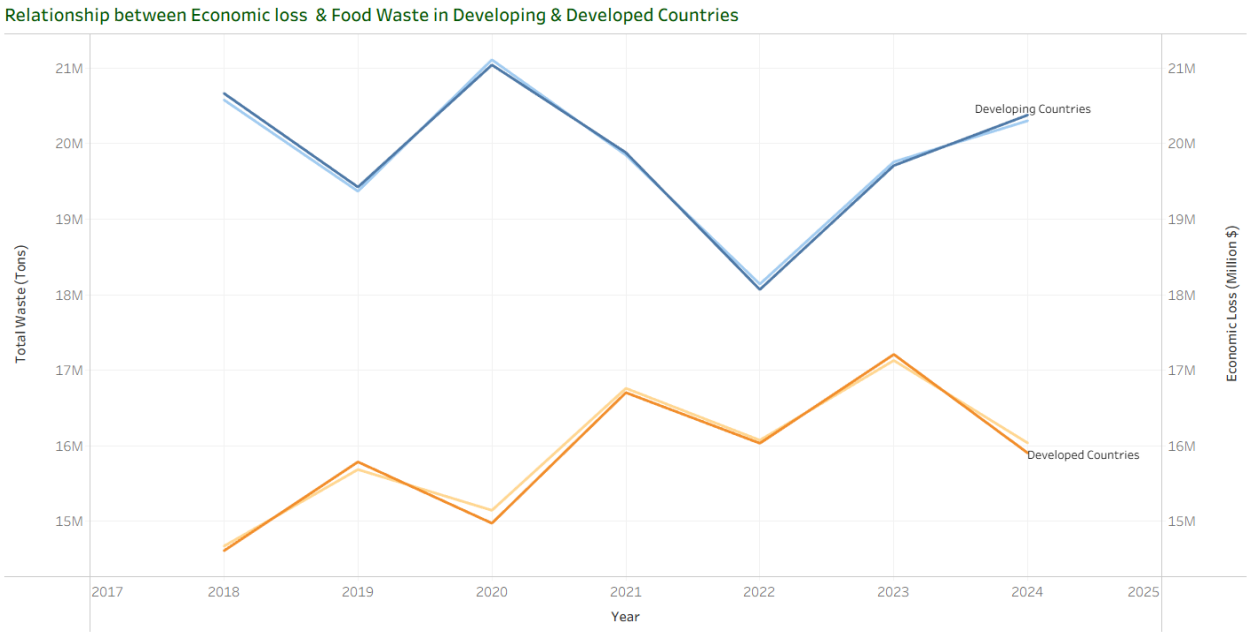
Graph 10: Map - Economic Loss (Millions$) in 20 Countries from 2018 to 2024



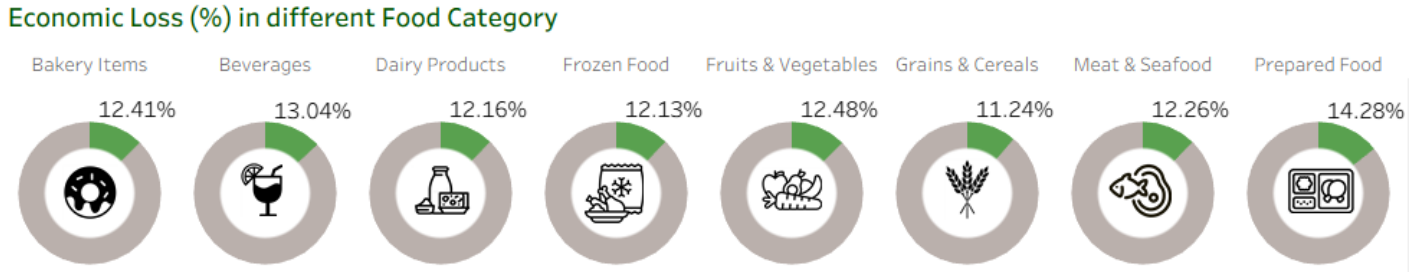
Graph 11: Treemap - Economic Loss(Million $) in 20 countries



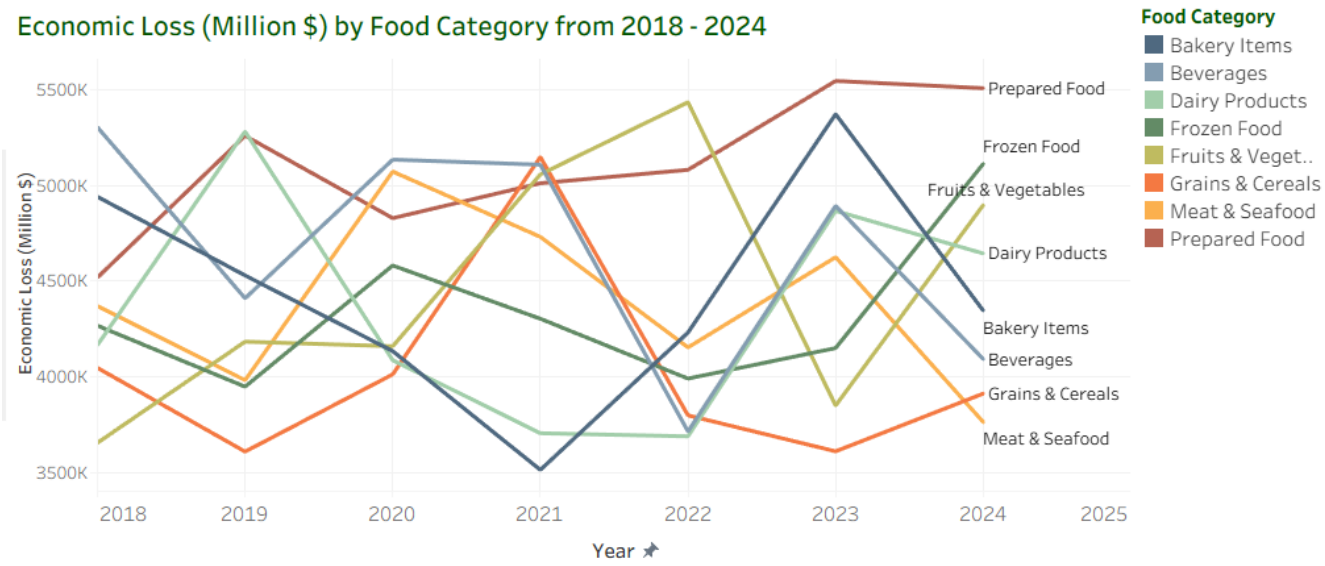
Graph 12: Line - Relationship between Economic Loss and Food Waste in Developing and Developed Countries



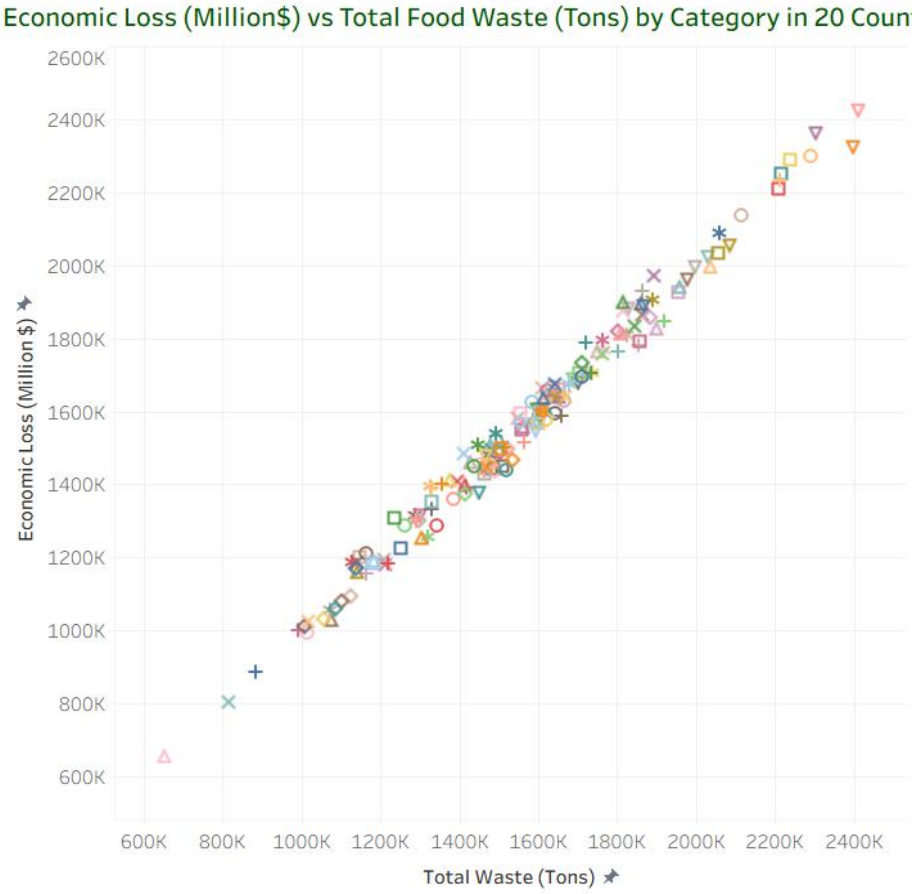
Graph 13: Donut Chart - Economic Loss (%) in different Food Categories



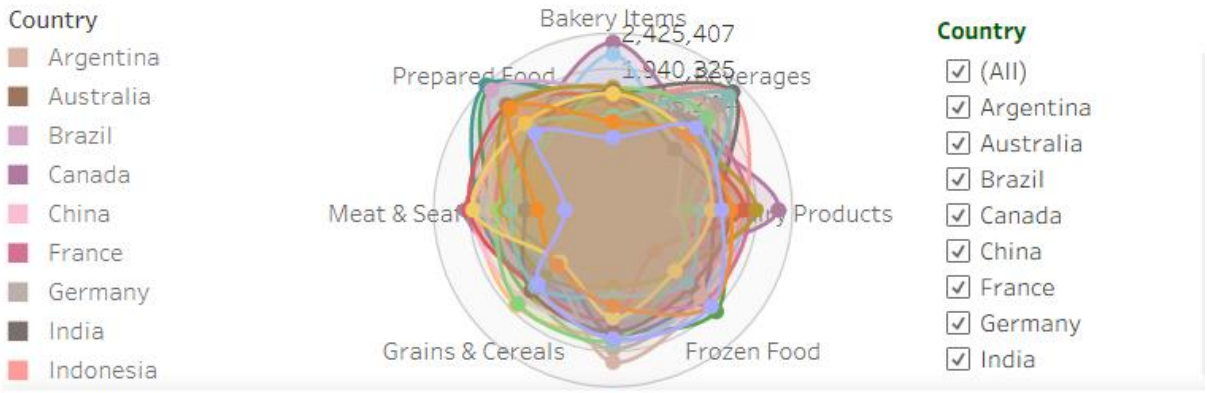
Graph 14: Line - Economic Loss (Million $) by Food Category from 2018 - 2024



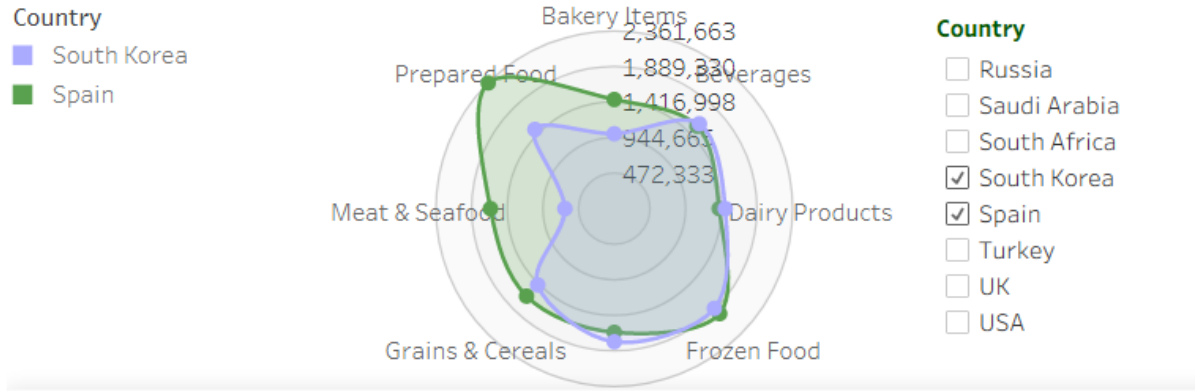
Graph 15: Scatter - Economic Loss (Million$) vs Total Food Waste (Tons) by Category in 20 Countries



Graph 16: Radar Chart - Economic Loss (Million $) in different Food Categories by 20 Countries



Economic Loss (Million $) in different Food Category by 20 Countries



Economic Loss (Million $) in different Food Catergory by 20 Countries